

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India 09/2021 – 01/2025 (3.5 Years)

- Analyzed and mitigated patient safety hazards in enterprise software by collaborating with cross-functional teams, reducing critical safety incidents by **15%**.
- Conducted comprehensive risk analyses for new software designs, assessing severity and occurrence for cardiac medical product features and evaluating risk–benefit trade-offs.
- Developed and maintained risk management files in compliance with **ISO 13485** and **ISO 14971**, ensuring alignment with design inputs and outputs for medical device software.
- Supported and executed reliability and safety engineering studies, including **Design of Experiments (DOE)**, to evaluate design concepts and system performance.
- Implemented design improvements and statistical process control techniques to enhance product reliability and safety, leading to a **20% reduction** in post-release defect rates.
- Provided data-driven risk–benefit insights and proposed risk mitigation strategies to stakeholders, prioritizing patient safety and regulatory compliance.

SKILLS

Safety Engineering – Risk Management, Hazard Analysis, Risk Analysis, PHA, HAL, FTA, DFMEA, Reliability Engineering, Design for Reliability (DfR), Design for Reliability and Manufacturability (DRM), Design for Six Sigma (DFSS), Design for Lean Sigma (DFLS), Robust Design Practices

Regulatory Compliance – ISO 13485, ISO 14971, IEC 62304, Medical Device Software, Global Regulatory Standards

Software Development – Java, Spring Boot, Python, C++, JavaScript, .NET, C#, REST APIs

Data Analysis & Visualization – Power BI, DAX, Tableau, R, Statistics, Data Modelling, Data Warehousing, DOE

Cloud & DevOps – AWS, Docker, Linux, Git, Jenkins, Agile Methodologies

System Engineering – Traceability, Design Verification, Validation, System Performance Evaluation, Change Management

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin 02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India 01/2021

PROJECTS

Medical Device Software Safety Analysis Platform

- Developed a Python-based application to automate the analysis of software logs for potential safety hazards in medical devices, integrating with **Splunk** data streams.

- Designed and implemented data models within **PostgreSQL** to store and query risk assessment data, facilitating compliance with **ISO 14971** standards for a hypothetical medical device.
- Created interactive dashboards using **Power BI** to visualize risk trends, failure modes, and mitigation effectiveness, enabling faster review by safety engineering teams.

Reliability Study Framework for Embedded Systems

- Engineered a Java Spring Boot microservice to manage and execute reliability testing protocols for embedded systems, ensuring adherence to **IEC 62304** requirements.
- Integrated with CI/CD pipelines using **Jenkins** to automate the deployment and testing of new reliability study configurations, reducing manual intervention by **40%**.
- Designed REST APIs for seamless integration with hardware test benches and data acquisition systems, providing real-time performance metrics.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)